[I225] Statistical Signal Processing(E) Office Hour 1

1. Three people are asked to choose a number from 1 to 10 randomly. Let's assume that each number have an equal probability, and each person cannot influence other people.

(1) What is the probability that all of the people are choosing the same number?
(2) What is the probability that at least 2 of them are choosing the same number?
(3) Probability that only two people are choosing the same number.

Answer:

(1)

We are tasked to find $P(A)$ = possibility where all the people are choosing the same number. To find the answer, there are 2 number we need to find:

- n_{all}: number of all possible cases = 10^3
- n_A: number of cases where each person is choosing the same number = 10

From this, we can get $P(same) = \frac{n_A}{n_{all}} = 0.01$

(2)

Let's first define the events.

B: Event where at least 2 people are choosing the same number.

Rather than finding $P(B)$, it is easier to find $P(B^C)$:

B^C is the opposite of event B, which is "event where every person is choosing a different number".

From $P(B^C)$ we can get

$$P(B) = 1 - P(B^C).$$

And, to get $P(B^C)$, we only need to number of favourable cases (n_B) and number of all possible cases (n_{all}):

- Number of favourable cases $= 10 \times 9 \times 8$

(Why $10 \times 9 \times 8$? Because, when person #1 has 10 number as his choice, person #2 need to choose from (10-1) numbers (10 possible numbers – 1 number that is chosen by person #1). It is also the case for person #3.)

- Number of all possible cases = 10^3

$$P(B^C) = \frac{n_B}{n_{all}} = 0.72$$

$$P(B) = 1 - P(B^C)$$

$$= 1 - 0.72$$

$$= 0.28$$

(3) Similar on how we answer subquestion B, we can use the opposite of event.

- C: event where only 2 people are choosing the same number.
- C^C: event where more or less than 2 people are choosing the same number
  → event where every person choose the same number OR every person choose different number → A U B^C

Since A and B^C are mutually exclusive,

$$P(A \cup B^C) = P(A) + P(B^C)$$

$$= 0.01 + 0.72$$

$$= 0.73$$

Therefore,

$$P(C) = 1 - P(A \cup B^C)$$

$$= 0.27$$

2. A pair of dice with 4 faces are rolled 5 times. What is the probability that from 5 enrollment, 2 of them have the same face?

Answer:

First, we need to find the probability of success case (p)

- p: probability that both of thrown dice has same face $=\frac{4}{4^2}=\frac{1}{4}$
- q: probability of failure $=1-p=\frac{3}{4}$

It is also known that:

- n: number of trials = 5
- m: desired number of successes = 2

Then, by using binomial distribution, we can find the answer:

$$X \sim Binomial\left(n=5, p=\frac{1}{4}\right)$$

$$P(X=2)=\binom{5}{2}\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^3$$

$$=10\cdot\frac{1}{16}\cdot\frac{27}{64}$$

$$=\frac{135}{512}$$

3. There are two boxes containing items of the same type.

- The first box contains 50 items, among which 10 are first-class items.
- The second box contains 30 items, among which 18 are first-class items.

A box is selected at random, and then two items are drawn from that box without replacement.

(1) Find the probability that the first item drawn is a first-class item.
(2) Given that the first item drawn is a first-class item, find the probability that the second item drawn is also a first-class item

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Answer:

Let's define the events:

- H denote the event that the first box is selected
- K denote the event that the second box is selected

Since the box is chosen randomly,

$$P(H) = P(K) = \frac{1}{2}$$

Let A_i denote the event that the i-th drawn items is a first-class item, i = 1,2.

(1) Probability that the first item is first-class:

$$P(A_1/H) = \frac{10}{50} = \frac{1}{5}$$

$$P(A_1/K) = \frac{18}{30} = \frac{3}{5}$$

Thus, by the law of total probability,

$$P(A_1) = P(A_1/H)P(H) + P(A_1/K)P(K) = \frac{1}{5} \times \frac{1}{2} + \frac{3}{5} \times \frac{1}{2} = \frac{2}{5}$$

(2) We need to compute:

Conditional probability $P(A_2/A_1)$

Using the law of total probability,

$$P(A_1A_2) = P(A_1A_2/H)P(H) + P(A_1A_2/K)P(K)$$

Under event H, two draws are made from the first box without replacement.

Thus,

$$P(A_1A_2/H) = \frac{10}{50} \times \frac{9}{49}$$

Similarly, under event K,

$$P(A_1A_2/K) = \frac{18}{30} \times \frac{17}{29}$$

Therefore,

$$P(A_2/A_1) = \frac{1}{P(A_1)}[P(A_1A_2/H)P(H) + P(A_1A_2/K)P(K)]$$

$$= \frac{5}{2}\left(\frac{10}{50}\cdot\frac{9}{49}\cdot\frac{1}{2} + \frac{18}{30}\cdot\frac{17}{29}\cdot\frac{1}{2}\right)$$

$$= \frac{1}{4}\left(\frac{9}{49} + \frac{51}{29}\right)$$

$$= 0.4856$$

4. A point (X, Y) is randomly and uniformly selected from the unit square region where $0 \le X \le 1$ and $0 \le Y \le 1$.

(1) Find the joint probability density function (PDF) of X and Y.
(2) Find the probability that the point lies within the triangle $T = \{(x, y) \mid 0 \le y \le x \le 1\}$.
(3) Find the conditional probability that $X > 0.5$ given that $Y < 0.5$.

Answer:

(1) Since the point is uniformly distributed over the unit square, the joint PDF is:

$$f(x,y) = \begin{cases} 1, & \text{if } 0 \le x \le 1 \text{ and } 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

(2) The probability that the point lies in triangle $T=\{(x,y): 0 \le y \le x \le 1\}$ is:

$$P\big((X,Y) \in T\big) = \iint_T f(x,y)\,dx\,dy = \int_0^1 \int_0^x 1\,dx\,dy = \int_0^1 x\,dx = \frac{1}{2}$$

(3) Conditional probability:

$$P(X>0.5 \mid Y<0.5) = \frac{P(X>0.5 \cap Y<0.5)}{P(Y<0.5)}$$

Numerator: Area of rectangle $[0.5,1] \times [0,0.5] = 0.5 \times 0.5 = 0.25$

Denominator: Area of $Y<0.5$ over the square $= 1 \times 0.5 = 0.5$

So,

$$P(X>0.5 \mid Y<0.5) = \frac{0.25}{0.5} = 0.5$$

5. Two people agree to meet at the east gate of a park sometime between 9:00 AM and 10:00 AM. The one who arrives first agrees to wait for at most 20 minutes (i.e., 1/3 hour). If the other person has not arrived by then, the first person will leave. What is the probability that the two people successfully meet?

Answer:

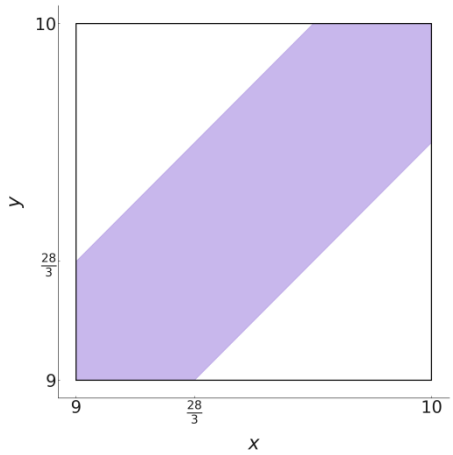
Let X and Y denote the arrival times of the two people.

Then X and Y are independent and uniformly distributed on the interval [0, 1].

- X~U(9,10) arrival time of person A
- Y~ U(9,10) arrival time of person B
- X and Y are independent

They will meet if the absolute difference between their arrival times is at most 1/3 hour, i.e.:

$$|X-Y| \leq \frac{1}{3}$$



So the desired probability is:

$$P(|X-Y|\leq 1/3) = \iint_{|x-y|\leq 1/3} f(x,y)dxdy$$

$$P(|X-Y|\leq 1/3) = \iint_{|x-y|\leq 1/3} dxdy$$

$$P(|X-Y|\leq 1/3) = 1-\iint_{|x-y|>1/3} dxdy$$

$$P(|X-Y|\leq 1/3) = 1-2\times\frac{1}{2}\left(\frac{2}{3}\right)^2$$

$$P(|X-Y|\leq 1/3) = \frac{5}{9}$$

6. Let X and Y be independent random variables. Suppose

$$X \sim B(n_1, p), \quad Y \sim B(n_2, p)$$

where B(n,p) denotes the binomial distribution.

Using characteristic functions, show that

$$X+Y \sim B(n_1+n_2, p).$$

Answer:

To prove that $X+Y$ follows a binomial distribution, we use characteristic functions.

Step 1: Characteristic function of a binomial random variable

If $Z \sim B(n,p)$, then its characteristic function is

$$\varphi_Z(t)=E[e^{itZ}]$$

Since a binomial random variable can be viewed as the sum of n independent Bernoulli random variables, its characteristic function is

$$\varphi_Z(t) = (1-p+pe^{it})^n$$

Therefore,

$$\varphi_X(t) = (1-p+pe^{it})^{n_1}$$

And

$$\varphi_Y(t) = (1-p+pe^{it})^{n_2}$$

Step 2: Use independence

Because X and Y are independent, the characteristic function of $X+Y$ is the product of the characteristic functions:

$$\varphi_{X+Y}(t)=\varphi_X(t)\varphi_Y(t)$$

Substituting the formulas above,

$$\varphi_{X+Y}(t) = (1-p+pe^{it})^{n_1}(1-p+pe^{it})^{n_2}$$

Thus,

$$\varphi_{X+Y}(t) = (1-p+pe^{it})^{n_1+n_2}$$

Step 3: Identify the distribution

But $(1-p+pe^{it})^{n_1+n_2}$ is exactly the characteristic function of a binomial random variable with parameters n_1+n_2 and p.

Hence, $X+Y \sim B(n_1+n_2, p)$.